No localizable particles in relativistic quantum theories

Hans Halvorson

November 12, 2013

Hegerfeldt's theorem

Suppose that $U_t = e^{itH}$, where H is a self-adjoint operator with half-bounded spectrum. Let A be a positive operator. Then for any unit vector ψ, either

$$\langle U_t\psi, AU_t\psi\rangle \neq 0, \qquad \text{for almost all } t \in \mathbb{R},$$

or

$$\langle U_t\psi, AU_t\psi\rangle = 0, \qquad \text{for all } t \in \mathbb{R}.$$

corollary: If $\langle U_t\psi, AU_t\psi\rangle = 0$ for t in an open interval no matter how small then it is constantly 0.

localization systems

A mapping $\Delta \mapsto E_\Delta$ from spatial regions to projection operators on $\mathcal{H}$.

A mapping $x \mapsto U(x)$ from the translation group on spacetime to unitary operators on $\mathcal{H}$.

translation covariance: for any Δ and for any translation x, $U(x)E_\Delta U(x)^* = E_{\Delta+x}$.

localization systems

localizability: if Δ and Δ' are disjoint regions of space at a fixed time then E_Δ and $E_{\Delta'}$ are orthogonal.

energy bounded below: for any timelike x, the infinitesimal generator H_x of the one-parameter group $t \mapsto U(tx)$ has spectrum bounded from below.

microcausality: if Δ and Δ' are disjoint, and if the distance between Δ and Δ' is nonzero, then for any timelike x, there is an $\epsilon > 0$ such that $[E_\Delta, E_{\Delta+tx}] = 0$ whenever $0 \leq t < \epsilon$.

Malament's theorem

Let $(H, \Delta \mapsto E_\Delta, x \mapsto U(x))$ be a localization system over Minkowski spacetime that satisfies:

1. translation covariance
2. localizability
3. microcausality
4. energy bounded below

Then $E_\Delta = 0$ for all Δ.

unsharp localization systems

spectral theorem: self-adjoint operators correspond to PV measures.

but: not all observables correspond to PV measures; unsharp observables correspond to POV measures.

An **unsharp localization system** consists of a POV measure $\Delta \mapsto A_\Delta$ and a unitary representation $x \mapsto U(x)$ of the translation group.

unsharp localization

localizability: If Δ and Δ' are disjoint spatial regions then $A_\Delta + A_{\Delta'} \leq I$.

theorem: If $(\Delta \mapsto A_\Delta, x \mapsto U(x))$ satisfies:

1. positive energy
2. localizability
3. microcausality

Then $A_\Delta = 0$ for all Δ.

local number operators

Let $\Delta \mapsto N_\Delta$ be a mapping from spatial regions to positive, possibly unbounded operators.

additivity: if Δ and Δ' are disjoint regions then $N_{\Delta\cup\Delta'} = N_\Delta + N_{\Delta'}$.

number conservation: if $\{\Delta_n : n \in \mathbb{N}\}$ is a disjoint covering of S, then the sum $\Sigma_n N_{\Delta_n}$ converges to a densely defined, self-adjoint operator N on $\mathcal{H}$, and $U(x)NU(x)^* = N$ for any timelike translation of x of M.

no local number operators

Suppose that the system $(\mathcal{H}, \Delta \mapsto N_\Delta, x \mapsto U(x))$ of local number operators satisfies:

1. additivity
2. translation covariance
3. energy bounded below
4. number conservation
5. microcausality

Then $N_\Delta = 0$ for all Δ.